normally generates very small reverse current. While in Zener diode, when the reverse voltage is increased to the breakdown voltage ($V_Z$), the increase in current is very sharp. The voltage remains almost constant throughout the breakdown region. In Figure 10.20(c), $I_{Z(max)}$ represents the maximum reverse current. If the reverse current is increased further, the diode will be damaged. The important parameters of the reverse characteristics are

- Zener breakdown voltage, $V_Z$
- Minimum current to sustain breakdown, $I_{Z(min)}$

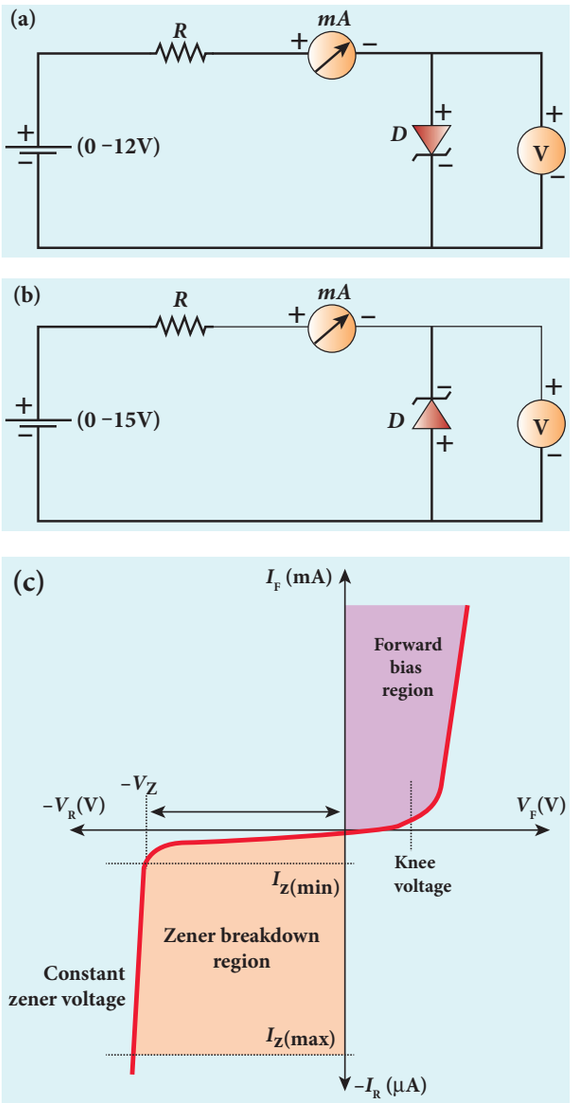



**Figure 10.20** Zener diode (a) Forward bias (b) Reverse bias (c) V-I characteristics

- Maximum current limited by maximum power dissipation, $I_{Z(max)}$

The Zener diode is operated in the reverse bias condition with the voltage greater than $V_Z$ and current less than $I_{Z(max)}$. The reverse characteristic is not exactly vertical which means that the diode possesses some small resistance called Zener dynamic impedance. Zener resistance is the inverse of the slope of the curve in the breakdown region. It means an increase in the Zener current produces only a very small increase in the reverse voltage. However this can be neglected. The voltage of an ideal Zener diode does not change once it goes into breakdown. In other words, **$V_Z$ remains almost constant even when $I_Z$ increases considerably**.

**Note**

The maximum reverse bias that can be applied before entering into the Zener region is called the peak inverse voltage, commercially referred as PIV rating.

## Applications

The zener diode can be used

- as voltage regulator
- for calibrating voltages
- to provide fixed reference voltage in a network for biasing
- to protect of any gadget against damage from accidental application of excessive voltage.

## Zener diode as a voltage regulator

Zener diode working in the breakdown region can serve as a voltage regulator whose circuit diagram is given in Figure 10.21. A series resistance $R_s$ of suitable value is used to limit the Zener current to avoid any damage to the diode. This resistance also plays a role in voltage regulation. The fluctuating DC



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